

OSCILLOSCOPE: Get The Best At The Best Price



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Are you planning to buy an oscilloscope? Pause and read this article first, because it might help you get complete value for your money. When you look at the latest generation of oscilloscopes, it is evident that a higher price does not necessarily mean higher performance or better features. This article aims to help you judge which factors really matter and which do not when buying an oscilloscope, and also points out some of the trendiest features in the latest products, so you can pick the right, budget-friendly oscilloscope for your needs.

With an oscilloscope, also known as scope, you can observe electrical signal waveforms such as amplitude, frequency, distortion and time intervals. Broadly, there are two types of oscilloscopes: analogue and digital.

One key difference between an analogue scope and a digital one is that the former uses a cathode ray tube. As compared to the digital variant, an analogue scope is generally less expensive. You can get an analogue scope that provides adequate performance and functions for use in many laboratories and service stations.

Budget-friendly oscilloscopes, generally speaking, are general-purpose entry-level oscilloscopes with good price-performance ratio. Such scopes are now becoming popular among hobbyists, students and experimenters. These are available in various form factors such as traditional benchtop, handheld, pocket and PC based. A few of these are handy instruments that you can easily carry in your toolbox or bag.

Currently, low-cost pocket-size oscilloscopes are mostly designed and manufactured in China, but are readily available on eBay and Amazon. A typical pocket-size oscilloscope is shown in Fig. 1 and a handheld oscilloscope in Fig. 2.

A PC based oscilloscope becomes a handy tool because you can observe the waveforms, like on any other oscilloscope,

Target audience

Target audience for budget-friendly oscilloscope is mostly small businesses, educationists and hobbyists, although some large companies also use these. Other target areas include universities and research institutes like IITs, NITs, IISERs, IITs, polytechnics, engineering and science colleges, skill development centres and schools, among others.

Analogue versus digital

Both types of oscilloscopes work differently but do almost the same thing—display and analyse input signals. Digital scopes have a much higher sample rate and larger memory, whereas analogue ones have the ability to adjust and focus the screen easily and are often much easier to use to analyse real-time signals.

Do not make cost your only priority. Check features like minimum and maximum amplitudes of signals, highest frequency of signals that you need to measure, storage requirements, multifunction capabilities, portability and printing requirement, among others. Ultimately, the choice depends on individual applications.

on your computer screen. Of course, it requires a specialised signal-acquisition board that normally uses a microcontroller or processor. The board is interfaced with a computer using an external USB or an internal add-on PCI or ISA card. The user interface and signal-processing software runs on the PC rather than on an embedded system, as in the case of a conventional scope.

A PC based oscilloscope is available at a much lower cost as compared to a conventional oscilloscope and is ideal for students. A typical PC based oscilloscope is shown in Fig. 3. Some commonly-available PC based oscilloscopes are listed in Table I.

Many high-bandwidth oscilloscopes have wide applications but cost much higher. However, high bandwidth is not always required. Some medium-range bandwidth (15MHz-100MHz) oscilloscopes are available for less than US\$ 300 as budget-friendly oscilloscopes from various sources.



Fig. 1: Typical pocket-size oscilloscope



Fig. 2: Typical handheld oscilloscope